COMSATS University Islamabad | Department of Computer Science

Operating Systems Lab — Mid

BS(AI-5) — 5th Semester | CSC322 | Total Marks: 50

Student Name: _____ Registration No.: _____

Q1. File and Directory Management

A university wants to organize student records using the Linux terminal. Create a sequence of Linux commands to:

1. Create a directory University containing subdirectories BSCS and BBA.
2. Inside BSCS, create students.txt and enter three student names.
3. Copy students.txt to BBA.
4. Rename the copied file to bba_students.txt.
5. Display the contents of both directories and save the BSCS directory listing to report.txt.

Q2. File Permissions and Text Processing

A company maintains employee records in employees.txt. Create a Linux command sequence to:

1. Create employees.txt and enter five employee names.
2. Display the names in alphabetical order.
3. Search for a specified employee by name.
4. Set owner permissions to read/write, group to read-only, and others to no access.
5. Save the sorted names to sorted_employees.txt and display the permissions of employees.txt.

Q3. Process Monitoring and Control

A laboratory administrator must manage a background application. Create a sequence of Linux commands to:

1. Display currently running processes.
2. Start a safe test process in the background.
3. Display the test process PID.
4. Terminate that test process using its PID.
5. Verify that the process has terminated.

Q4. Parent and Child Processes Using fork()

A software company needs a program demonstrating parent-child process creation. Create a C++ program to:

1. Create one child process using fork().
2. Display the parent process ID.
3. Display the child PID and parent PID from the child process.
4. Print “Child Process Executing” from the child.
5. Print “Parent Process Executing” from the parent; compile and execute in Ubuntu. Make the parent wait using wait().

Question 5: Text Processing and Pipelining in Linux

Scenario: A university maintains a file named students.txt containing student records. The administrator wants to search, sort, and examine these records efficiently using Linux commands.

Create a sequence of Linux commands to perform the following operations:

1. Create students.txt containing at least ten student names.
2. Use grep to search for names containing the letter a.
3. Use sort to arrange the student names alphabetically.
4. Use head and tail to display the first five and last five records.
5. Use a pipeline (|) to search for names containing a, sort the matching names alphabetically, and display only the first three results.

Q5 Student Result Processing

A university wants to automate student results. Create a Bash script named student_result.sh to:

1. Read a student name and marks in three subjects.

2. Calculate total marks.

3. Calculate percentage, assuming each subject has 100 marks.

4. Print Pass for percentage ≥ 50%, otherwise Fail.

5. Display student name, total, percentage and result.

Q6. Automated File Management

A company needs a daily report generator. Create a Bash script named file_manager.sh to:

1. Create Daily_Reports if it does not already exist.

2. Create report1.txt, report2.txt and report3.txt in that directory.

3. Write a different message into each file.

4. Combine the three files into final_report.txt and display the result.